

UAE Regulatory Compliance

Legal alignment mapping for the GOYA Ledger platform against United Arab Emirates regulation.

Covers: electronic signatures, blockchain/DLT, digital identity, data protection, and smart contracts.

Disclaimer: This document is a **self-assessment** mapping code features to legal requirements. It is not a certification, legal opinion, or formal audit result. No third-party auditor, TDRA, or UAE legal authority has validated these claims. Organizations should seek independent legal counsel before relying on these mappings for compliance purposes.

Federal Decree-Law No. 46/2021 — Electronic Transactions and Trust Services

Replaced Federal Law No. 1/2006. Effective February 2022.

Art. 1 — Definitions

Legal concept	GOYA implementation	Status
Electronic signature: letters, numbers, symbols, voice, or process in electronic form associated with a data message	SignatureLevel::Simple — Ed25519 DID-based signature over content hash	Done
Advanced electronic signature: uniquely linked to signatory, identifies signatory, under sole control, detects subsequent changes	SignatureLevel::Advanced — ML-DSA-65 + biometric commitment	Done
Qualified electronic signature: advanced signature created with QSCD, based on	SignatureLevel::Qualified — reserved for QTSP integration	Planned

Legal concept	GOYA implementation	Status
qualified certificate from TDRA-licensed QTSP		

Art. 11 — Automated Electronic Agents (Smart Contracts)

Requirement	GOYA compliance
A contract formed by interaction of automated electronic agents is valid, enforceable, and legally effective even if no natural person directly intervened	Governance proposals, voting, and inference claims execute via deterministic on-chain logic. Chaincode module provides programmable contract execution
Applies to interactions between two automated agents and between an agent and a natural person	API-driven transactions (automated) and wallet-signed transactions (natural person) both produce legally valid on-chain records

Art. 19 — Advanced Electronic Signature Requirements

Requirement	GOYA control	Evidence
(a) Uniquely linked to the signatory	DID (did:goya:{pubkey}) + biometric commitment(s) hashed into signing payload	compute_biometrics_hash() bound to payload
(b) Capable of identifying the signatory	Biometric evidence types: fingerprint, facial recognition, Emirates ID, iris, voice, government ID. Caveat: identification relies on self-issued DIDs — full Art. 19(b) compliance requires UAE Pass or Emirates ID verification via RA	BiometricType enum, ProofingMethod::UaePass
(c) Created using data under sole control of the signatory	ML-DSA-65 private key generated client-side, encrypted with Argon2id + AES-256-GCM	Client-side key management
(d) Linked to data so that any change is detectable	PQC signature covers signer DID + content hash + biometric hash	Signing payload: "notarize_fea:{s}:{h}:{bio_hash}"

Art. 20 — Qualified Electronic Signature

Requirement	Status
Based on a qualified certificate issued by a TDRA-licensed QTSP	Planned — Qualified level reserved for TDRA-licensed provider integration
Created using a QSCD (Qualified Signature Creation Device)	HsmSigningProvider (PKCS#11) supports HSM-backed signing. Production deployment requires Common Criteria EAL4+ certified HSM

Art. 21 — QSCD Requirements

Requirement	GOYA readiness
Confidentiality of signature creation data	HSM keeps private key in hardware; SimulatedHsmProvider enforces session-gated access
Protection against use by third parties	PIN/session authentication on HsmSigningProvider
Signature created once per operation	Atomic signing operation in SigningProvider::sign()
Data to be signed not altered before signing	Signing payload constructed from immutable inputs, signed in single operation

Art. 10 — Record Retention

Requirement	GOYA implementation
Electronic records must be retained for the period specified by law (15 years for trust services)	UAE_RETENTION_SECS constant (15 years). AuditRetentionPolicy::with_min_retention_years(15) configurable per deployment
Records must be accessible and reproducible	AuditStore::export_json() with hash chain metadata. RocksDB persistent storage with tamper-evident audit chain

Cabinet Resolution No. 28/2023 — Executive Regulation

Implements DL 46/2021 with technical requirements.

Requirement	GOYA compliance
TDRA licensing for trust service providers	Operational requirement — code supports QTSP certificate chain validation when provider is licensed
ETSI standards adoption	CAdES (ETSI TS 101 733), XAdES (ETSI TS 101 903), PAdES (ETSI TS 102 778), TSA (RFC 3161), OCSP (RFC 6960), CRL (RFC 5280) all implemented
Biennial conformity assessment (Art. 35-36)	Audit trail infrastructure supports assessment: tamper-evident hash chain, JSON/CSV export, retention policy
Mandatory security breach notification	Audit event categories include security officer actions, system events, and incident-relevant entries (AuditAction enum)

TDRA Trust Service Provider Requirements

TSP service (Art. 17)	GOYA module	Status
Electronic signature creation	SigningProvider trait (Ed25519, ML-DSA-65, RSA)	Done
Qualified certificate issuance	CaHierarchy (root + intermediate CA), pki.rs	Done
QSCD management	HsmSigningProvider (PKCS#11, cryptoki)	Done
Electronic preservation	NotarizationEntry on-chain with block height anchoring	Done
Validation services	OCSP responder (RFC 6960), CRL distribution (RFC 5280)	Done
Electronic seals	SignatureLevel::Seal for legal entity signatures	Done
Time stamping	TsaProvider (RFC 3161), DER-encoded timestamps	Done

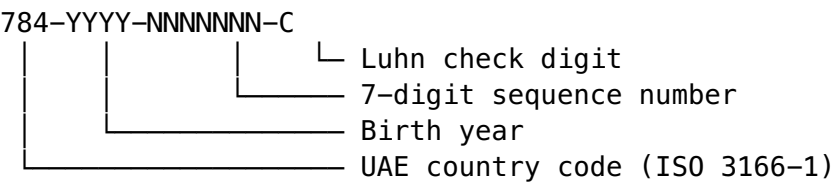
TSP service (Art. 17)	GOYA module	Status
Qualified electronic delivery	Not implemented	Planned

Identity — Emirates ID and UAE Pass

Registration Authority Integration

Component	Implementation
Emirates ID validation	<code>validate_emirates_id()</code> — format 784-YYYY-NNNNNNN-C, Luhn check digit
UAE Pass verification method	<code>ProofingMethod::UaePass</code> in RA module
Jurisdiction routing	<code>Jurisdiction::Uae</code> dispatches to Emirates ID validation via <code>validate_national_id()</code>
IdentityProofing struct	<code>national_id</code> field for Emirates ID, <code>jurisdiction</code> field for multi-jurisdiction support

Emirates ID Format



Validation: country code 784, year range 1900-2100, Luhn-10 check digit.

Data Protection — PDPL (Federal Decree-Law No. 45/2021)

Requirement	GOYA compliance
Lawful basis for processing personal data	Application layer responsibility. Biometric commitments are SHA-256 hashes (derived data, not raw biometric)
Cross-border data transfer requires adequate protection or contractual safeguards	Deployment configuration — node can be deployed within UAE for data residency. No code-level enforcement

Requirement	GOYA compliance
Data subject rights (access, correction, deletion)	DID-based identity allows subject identification. Deletion constrained by blockchain immutability and 15-year retention requirement
CBUAE requirement: financial data stored within UAE	Deployment constraint — RocksDB data directory configurable via GOYA_DATA_DIR

Biometric Data Under PDPL

Aspect	Assessment
Classification	SHA-256 commitments are derived data. Conservative treatment: sensitive personal data
Storage	Commitment-only architecture — no raw biometric data enters the system
Consent	Required for biometric processing. Application layer responsibility
Purpose limitation	Commitments used exclusively for signature binding

DLT/Blockchain Regulatory Landscape

VARA — Dubai (Law No. 4/2022)

Requirement	Relevance to GOYA
VASP licensing for virtual asset activities	Applies if GOYA tokens are classified as virtual assets in Dubai
4 mandatory rulebooks + 7 activity-specific	Compliance depends on deployment use case (custody, exchange, etc.)
AML/CFT compliance	KYC via RA module (Emirates ID + UAE Pass). Travel Rule integration is application-layer concern

ADGM — Abu Dhabi (DLT Foundations Regulations 2023)

Requirement	Relevance to GOYA
First DLT Foundations and DAO regime worldwide	GOYA governance module (proposals, voting, quorum) aligns with DAO governance requirements
FSRA licensing for financial services on DLT	Required if GOYA deployed for regulated financial services in ADGM

Federal — SCA and CBUAE

Regulator	Scope	GOYA relevance
SCA	Virtual assets at federal level	Token classification determines applicability
CBUAE	Payment tokens (Payment Token Services Regulation 2024)	Applies if GOYA token used as payment instrument

Cryptographic Alignment

Requirement	GOYA implementation
Technology-neutral (no algorithm mandates)	Ed25519, ML-DSA-65 (FIPS 204), RSA-2048 supported
ETSI-referenced standards (implicit RSA 2048+ or ECDSA P-256+)	RSA-2048 PKCS#1 v1.5 via RsaSigningProvider. Ed25519 exceeds ECDSA P-256 security level
HSM for QSCD (Common Criteria EAL4+)	HsmSigningProvider via PKCS#11/ cryptoki. Production requires certified HSM hardware
Post-quantum readiness	ML-DSA-65 deployed — UAE law does not prohibit PQC algorithms

Language and Localization

Requirement	Status
Arabic language support in interfaces and RTL	Application/frontend concern. API responses are JSON (language-neutral)
Arabic or bilingual documents	Certificate policies (CertificatePolicy, CPS) support configurable text. Arabic content requires deployment configuration
2023 amendments reinforce Arabic support	Tauri desktop app and web frontends would need Arabic/RTL UI layer

Jurisdictional Comparison

Feature	Chile (Ley 19.799)	EU (eIDAS)	UAE (DL 46/2021)	GOYA
Signature tiers	2 (FES/FEA)	3 (simple/advanced/qualified)	3 (simple/advanced/qualified)	3 (Simple/Advanced/Qualified)
Qualified requires TSP	PSC acreditado	QTSP in EU Trusted List	QTSP licensed by TDRA	Qualified reserved for TSP integration
National ID	RUT (modulo 11)	Per member state	Emirates ID (Luhn)	validate_national_id() dispatches by Jurisdiction
Identity proofing	Ley 19.799 Art. 15	eIDAS Art. 24	DL 46/2021 + UAE Pass	RaStore with ProofingMethod variants
Retention period	7 years (DS 181)	Varies by member state	15 years (Art. 10)	Configurable: DEFAULT_RETENTION_SECS / UAE_RETENTION_SECS
Smart contract recognition	No explicit law	No explicit recognition	Art. 11 — explicitly valid	Governance + chaincode modules
Data residency	No explicit mandate	GDPR (adequate protection)	PDPL + CBUAE mandate	Deployment configuration
PQC stance	No position	eIDAS 2.0 recommends readiness	Technology-neutral	ML-DSA-65 deployed

Gaps and Roadmap

Gap	Severity	Path to close
TDRA QTSP licensing	Blocker for Qualified	Legal/operational — apply to TDRA when ready
UAE Pass integration (live API)	Medium	Integrate UAE Pass OAuth/OIDC for real-time identity verification
Arabic UI/RTL	Medium	Frontend localization layer in Tauri app and web interfaces
CBUAE data residency enforcement	Low	Deployment documentation for UAE-hosted nodes
VARA/ADGM licensing	Conditional	

Gap	Severity	Path to close
		Depends on token classification and deployment jurisdiction
Qualified electronic delivery	Low	Not core to DLT use case

References

- [Federal Decree-Law No. 46/2021 — Electronic Transactions and Trust Services](#)
- [Cabinet Resolution No. 28/2023 — Executive Regulation](#)
- [TDRA — Trust Services Laws and Regulations](#)
- [Federal Decree-Law No. 45/2021 — Personal Data Protection Law](#)
- [Law No. 4/2022 — VARA \(Dubai\)](#)
- [ADGM DLT Foundations Regulations 2023](#)
- [CBUAE Payment Token Services Regulation 2024](#)